

ICE COFFEE "C" FUTURES (KC)

Spread & Curve Structure — Fundamental Reference

Brazil, Vietnam, Frost and Drought, Arabica-Robusta, and the Forward Curve

This document presents a fact-based reference on the fundamental drivers of ICE Coffee "C" futures (ticker: KC), the global benchmark for washed and unwashed Arabica coffee traded on the Intercontinental Exchange (ICE Futures US, formerly NYBOT). Coffee is a tree crop with a multi-year production cycle, concentrated in a relatively small number of tropical and subtropical producing countries, and subject to a biennial yield pattern (the "on-year/off-year" cycle) that is unique among major agricultural futures markets. Brazil dominates as both the largest Arabica producer and the country whose weather risk — frost in its southern growing regions, a genuinely unusual risk for a tropical crop — has produced some of the most dramatic price spikes in agricultural futures history. Coffee also features a fundamental species divide between Arabica (priced on KC) and Robusta (priced on ICE Europe's Robusta contract), with the relative price and availability of each driving substitution in blending and instant coffee production. Understanding the Brazilian crop cycle, the growing role of Vietnam and other Robusta producers, certified stock dynamics, and the consumption-side roasting and retail chain is the foundation of spread and butterfly trading on the KC curve.

Futures First — Internal Education Material
For Analyst Group Use

1. Contract Specifications and Delivery Mechanics

ICE Coffee C is a physically delivered contract for washed and unwashed Arabica coffee. The grade and origin differential structure is unusually elaborate compared to other soft commodities, reflecting the wide diversity of growing origins and processing methods that produce deliverable Arabica.

Parameter	Detail
Exchange	ICE Futures US (formerly NYBOT — "Coffee, Sugar & Cocoa Exchange," CSCE)
Ticker	KC (front month); commonly referred to as "Coffee C" or simply "C"
Underlying	Washed Arabica coffee (<i>Coffea arabica</i>), green (unroasted) beans
Contract size	37,500 pounds (approximately 250 bags of 60 kg, or roughly 17 metric tonnes)
Quotation	US cents per pound (¢/lb); minimum tick = 0.05¢/lb = \$18.75 per contract
Contract months	March (H), May (K), July (N), September (U), December (Z) — five delivery months per year. Liquid trading concentrated in front 3–4 contracts; December is typically the most actively traded as the post-Brazilian-harvest contract.
Settlement type	Physical delivery via warehouse receipt at ICE-licensed warehouses in the US (and select ports), graded and certified by ICE-approved graders.
Delivery grade	Growths from approved origins are assigned a differential (premium or discount) to the base contract grade. Colombian Excelso and other premium washed Arabicas earn a premium; lower-grade or less-preferred origins are discounted. Cup quality (taste defects), bean defect count, and moisture content all factor into the grading and differential assignment.
Last trading day	Approximately 8–9 business days prior to the last business day of the delivery month
Trading hours	ICE electronic: Sunday–Friday 21:15–14:00 ET (approx.); most liquid 09:00–13:30 ET
Contract value at \$2.00/lb	\$75,000 per contract
Relationship to Robusta	ICE Futures Europe Robusta Coffee (RC, London): the global benchmark for Robusta. Quoted in USD/tonne. The Arabica-Robusta spread is a key structural relationship reflecting relative quality premium and blending/instant-coffee substitution economics.

1.1 The Differential System — Why Origin Matters for Delivery

- Unlike many agricultural contracts with a single, narrowly defined deliverable grade, Coffee C allows delivery of coffee from approximately 20+ approved origins (Brazil, Colombia, Central America, parts of Africa and Asia), each assigned a differential — a premium or discount in cents per pound relative to the base contract grade — reviewed and adjusted periodically by the exchange based on quality and market assessments.
- **Washed vs natural (unwashed/dry) processing:** washed (wet-processed) Arabica generally commands a quality premium over natural (dry-processed) Arabica, as the washing process tends to produce more consistent cup quality. Brazil produces substantial quantities of both processing types, while Colombia and most Central American origins are predominantly washed.
- **The differential structure as a basis signal:** shifts in relative origin differentials (for example, Colombian premiums widening relative to Brazilian Santos-type coffee) reflect changes in relative

quality availability and regional supply conditions, and are tracked by trade houses as a basis-level signal distinct from the flat-price direction of the futures contract itself.

1.2 Arabica vs Robusta — The Fundamental Species Divide

- **Arabica (*Coffea arabica*):** grown at higher altitudes (typically 600–2,000+ metres), in cooler, more temperate tropical climates, producing a smoother, more aromatic cup generally used in specialty, espresso, and premium roast-and-ground coffee. More susceptible to disease (notably coffee leaf rust) and frost damage (in Brazil's southern growing areas) than Robusta.
- **Robusta (*Coffea canephora*):** grown at lower altitudes in hotter, more humid climates, higher caffeine content, more bitter and less aromatic cup, used predominantly in instant coffee, lower-cost blends, and (increasingly) some espresso blends for crema and bitterness. More disease-resistant and higher-yielding per hectare than Arabica, making it generally lower-cost to produce.
- **The Arabica-Robusta price spread and substitution:** roasters and instant coffee manufacturers can, within limits, adjust the Arabica-Robusta blend ratio in their products in response to relative prices. When the Arabica premium over Robusta widens significantly, manufacturers — particularly of instant coffee and lower-cost blends — shift toward higher Robusta content, moderating Arabica demand at the margin. This substitution operates with a meaningful lag (reformulation and consumer acceptance take time) and is bounded by what consumers will accept in terms of taste change.

2. Global Production — The Supply Landscape

World coffee production (Arabica plus Robusta combined) is approximately 170–180 million 60-kg bags per year, of which Arabica represents roughly 55–60% and Robusta the remainder. The coffee marketing year runs October–September (aligned with the Brazilian harvest cycle), the convention used by the USDA and the ICO (International Coffee Organization) for global balance sheet reporting.

Country	Production (approx.)	Species	Harvest Period	Key Risk	KC Curve Impact
Brazil	55–65 M bags	~75% Arabica / 25% Robusta (Conilon)	Apr–Sep (Arabica); year-round picking in some regions	Frost (Jul–Aug, southern Minas Gerais/Parana — rare but historically extreme); drought (flowering Sep–Oct, critical for the following year's crop); biennial bearing cycle	By far the dominant supply variable. Brazilian frost and drought events have produced the largest single-day and multi-week KC price moves on record.
Vietnam	27–32 M bags	~95%+ Robusta	Oct–Jan harvest (Central Highlands — Dak Lak province dominant)	Drought (linked to El Nino); ENSO-driven rainfall variability; farmer switching to other crops (durian, pepper) during low Robusta price periods	Not directly deliverable against KC (Robusta), but Vietnamese supply conditions drive the Robusta price and therefore the Arabica–Robusta spread and substitution dynamics.
Colombia	11–14 M bags	100% Arabica (washed)	Two harvest cycles per year (main crop Oct–Dec, "mitaca" secondary crop Apr–Jun) due to the bimodal rainfall pattern	La Nina: excess rainfall, leaf rust (roya) outbreak risk, road/logistics disruption; El Nino: drought stress	Premium washed Arabica; Colombian supply disruptions widen the premium for quality washed coffee relative to Brazilian natural-process Arabica.
Indonesia	10–12 M bags	~75% Robusta / 25% Arabica	Apr–Sep (varies by island — Sumatra, Java, Sulawesi)	ENSO-driven rainfall variability; smallholder farming structure limits rapid supply response	Significant Robusta producer; relevant for the Arabica–Robusta spread and for instant coffee supply chains.
Ethiopia	7–8 M bags	100% Arabica (wild and cultivated heirloom varieties)	Oct–Jan harvest	Smallholder/wild-forest production structure; limited large-scale weather data infrastructure; domestic consumption absorbs a substantial share of production	The historical genetic origin of Arabica; specialty-grade premium origin; smaller volume impact on the flat-price level but relevant for specialty differentials.
Honduras, Peru, Guatemala, others (Central/South America)	20–25 M bags combined	Predominantly Arabica (washed)	Oct–Mar (varies by country and altitude)	Leaf rust (roya) — the 2012–2013 Central American roya epidemic was severe; hurricane risk (Honduras, Guatemala); smallholder farmer economics and migration	Collectively significant washed Arabica supply; roya outbreaks have historically reduced regional production for multiple consecutive years.

India, Uganda, other Robusta origins	10–12 M bags combined	Predominantly Robusta	Varies	Monsoon variability (India); regional political/logistics factors (Uganda)	Secondary Robusta supply contributing to the global Robusta balance and Arabica-Robusta spread dynamics.
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2.1 The Biennial Bearing Cycle — A Defining Structural Feature

- **What biennial bearing is:** Arabica coffee trees (and to a lesser extent Robusta) have a natural tendency to produce a heavy crop ("on-year") followed by a lighter crop ("off-year"), as the tree depletes its carbohydrate reserves producing a large crop and requires a season of lower yield to replenish them before the next heavy crop. This pattern is most pronounced in Brazil, where it is large enough to be visible at the national production level, and is a key reason Brazilian crop estimates must always be interpreted relative to where the country sits in its on-year/off-year cycle, not in isolation.
- **Interaction with weather:** a weather-related yield shock (frost or drought) landing in an already-scheduled off-year compounds the production shortfall beyond what either factor would produce alone — a combination that has historically preceded some of the most significant KC price rallies. Conversely, favourable weather in an on-year can produce an exceptionally large crop that weighs on prices for an extended period.

3. Brazil — The Dominant Supply Variable

Brazil's role in the global coffee market is analogous to Brazil's role in sugar — a single country whose production cycle, weather risk, and currency dynamics dominate the global price-setting process. Brazil is the world's largest coffee producer and exporter and the only major coffee-growing country genuinely exposed to frost risk, a fact that has produced some of the most dramatic price moves in the history of agricultural futures.

3.1 Brazilian Growing Regions

- **Minas Gerais:** the dominant Arabica-producing state, accounting for roughly half of Brazilian Arabica output, including the prized Cerrado, Sul de Minas, and Mogiana growing areas. Minas Gerais' southern reaches are also the area most exposed to the rare but severe frost events discussed in Section 3.2.
- **Sao Paulo and Parana:** historically significant Arabica regions, with Parana having suffered the most catastrophic documented Brazilian frost damage historically (the 1975 "black frost," discussed below), though Arabica cultivation has since shifted further north and to higher elevations partly in response to recurring frost risk in Parana.
- **Espirito Santo and Bahia:** major Robusta (locally called "Conilon") producing regions, contributing to Brazil's significant Robusta output alongside its dominant Arabica production.

3.2 Frost Risk — A Genuinely Unusual Tropical Crop Hazard

- **Why frost is possible in a "tropical" crop:** Brazil's southern coffee-growing regions, while subtropical rather than equatorial, are occasionally affected by polar air masses pushing unusually far north during the Southern Hemisphere winter (June–August), bringing temperatures low enough to damage or kill coffee trees. This is a distinctive risk among major tropical commodities and is the single most powerful acute bullish catalyst in the coffee market.
- **The 1975 "Black Frost":** the most catastrophic documented Brazilian coffee frost event, affecting Parana state in July 1975 and destroying a very large share of the standing crop and a substantial number of trees outright (unlike a drought, which reduces yield, a severe enough frost kills the tree itself, requiring multi-year replanting and regrowth before full production capacity is restored). This event produced a multi-year supply disruption and a historic price spike, and remains the reference event against which subsequent Brazilian frost scares are measured.
- **Subsequent documented frost events:** Brazil has experienced several further frost events of varying severity in the decades since 1975 (including events in the 1990s and a notable frost scare in July 2021 affecting Minas Gerais and Parana that contributed to a sharp KC rally), though none has matched the 1975 event in scale. The geographic shift of Arabica cultivation further north and to higher elevations since the 1970s has reduced (but not eliminated) the crop's frost exposure.
- **The market mechanism:** because frost kills trees rather than merely reducing a single season's yield, the market reaction to a confirmed severe frost event prices in a multi-year reduction in productive capacity, not just a one-season shortfall — which is why frost scares can produce price reactions disproportionate to the immediately quantifiable crop damage, particularly in the hours and days before the actual extent of damage can be surveyed and confirmed.

3.3 Drought and the Flowering Period

- **The September–October flowering window:** Brazilian Arabica trees flower in September–October, and adequate soil moisture during this period is critical for the following year's crop (i.e., a drought in Sep–Oct 2026 would affect the 2027 harvest). This creates a distinctive forward-looking weather risk window: a dry flowering period is a bullish signal for the harvest more than a year away, not for the imminent harvest.

- **Cherry development drought risk:** drought stress during the cherry development period (roughly December–March) can reduce bean size and yield for that season's harvest more directly, a separate and more immediate weather risk window from the flowering-period risk described above.
- **CONAB Brazilian crop estimates:** Brazil's National Supply Company (CONAB) publishes coffee production estimates several times per year, with increasing precision as the harvest approaches. The first-half-year estimates (which must account for both the biennial cycle position and weather-to-date) are watched closely but carry wider uncertainty bands than the estimates published closer to harvest.

3.4 Brazilian Real (BRL/USD) and Farmer Selling Behaviour

- As with Brazilian sugar (and for the same underlying reason — global USD pricing against BRL-denominated production costs), BRL depreciation improves the BRL-equivalent revenue Brazilian coffee farmers receive at any given USD price, incentivising farmer selling and export, a structurally bearish effect on KC at the margin. BRL appreciation has the opposite effect, slowing farmer selling and supporting prices.
- **Farmer selling pace as a tracked variable:** Brazilian coffee farmers (and the cooperatives that aggregate and market much of the crop) have some discretion over the timing of physical sales within a marketing year, and the pace of farmer selling — tracked through export and domestic trade data — is itself a market signal, as slow farmer selling (often associated with BRL strength or a bullish price outlook among growers) can tighten the available exportable supply even when the underlying crop size is adequate.

4. Vietnam and the Robusta Supply Chain

Vietnam is the world's largest Robusta producer and the second-largest coffee producer overall, a position achieved through a rapid, policy-supported expansion of coffee cultivation since the late 1980s/1990s. While Vietnamese coffee is not directly deliverable against the Arabica-based KC contract, Vietnamese supply conditions are central to the global Robusta balance and therefore to the Arabica-Robusta spread that drives blending and substitution dynamics.

4.1 Vietnamese Production Structure

- **Central Highlands dominance:** the Dak Lak, Lam Dong, Gia Lai, and Dak Nong provinces in Vietnam's Central Highlands account for the substantial majority of national Robusta production, grown predominantly by smallholder farmers on relatively small plots.
- **Harvest and crop calendar:** the Vietnamese harvest runs approximately October-January, with the bulk of export volume reaching the market in the months following harvest completion.
- **ENSO sensitivity:** Vietnamese Robusta production, like much of mainland Southeast Asian agriculture, is sensitive to ENSO-driven rainfall variability — El Nino years are historically associated with drought stress and reduced Vietnamese yields, while La Nina years tend to bring more favourable rainfall.
- **Crop switching risk:** during periods of sustained low Robusta prices, Vietnamese smallholders have at times shifted land toward alternative crops (durian, black pepper, and others) that offered better relative returns, a structural supply-side response that operates over a multi-year horizon as replanting decisions take years to fully reverse.

4.2 The Vietnamese Domestic Marketing System

- Vietnamese coffee marketing involves a network of domestic traders and exporters (including significant participation by multinational trading houses) purchasing from smallholders and aggregating volume for export. Vietnamese farmer and trader selling pace, tracked via export statistics published by Vietnam Customs and industry associations, is a closely watched indicator of available near-term Robusta supply to the global market.

4.3 Robusta's Growing Role in Specialty and Espresso

- While historically associated primarily with instant coffee and lower-cost blends, improved Robusta processing and quality-focused cultivation (notably "fine Robusta" initiatives) have expanded Robusta's role in some espresso blends (traditionally, Italian espresso blends often include a Robusta component for crema and body) and even some specialty-positioned products. This is a slow-moving demand-side development that modestly increases the addressable market for Robusta beyond its traditional instant-coffee base.

5. Other Key Producers and Coffee Leaf Rust

5.1 Colombia

- Colombia's bimodal rainfall pattern (driven by the country's equatorial location and Andean topography) produces two harvests per year in most growing regions — the main crop (October–December) and a secondary "mitaca" or "fly" crop (April–June) — a structure distinct from Brazil's single annual Arabica harvest. The Colombian Coffee Growers Federation (FNC) publishes monthly production and export data that is closely tracked as the primary source for Colombian supply conditions.
- **La Nina and leaf rust risk in Colombia:** La Nina-associated excess rainfall in Colombia has historically been linked to coffee leaf rust (roya) outbreaks, as the fungus thrives in humid conditions — a notable 2008–2011 period of La Nina-linked excess rainfall and rust pressure significantly reduced Colombian production for several consecutive years before recovery aided by rust-resistant variety replanting programmes.

5.2 Coffee Leaf Rust (Roya) — The Primary Disease Risk

- **What it is:** *Hemileia vastatrix*, a fungal pathogen affecting Arabica coffee, causes leaf lesions that reduce the plant's photosynthetic capacity and, in severe outbreaks, can substantially reduce yield and in the most severe cases contribute to tree decline. Robusta is generally more rust-resistant than Arabica, making rust predominantly an Arabica-specific risk.
- **The 2012–2013 Central American epidemic:** a severe and widespread roya outbreak across Central America and parts of southern Mexico in 2012–2013, linked by researchers to a combination of favourable fungal conditions and ageing, rust-susceptible tree stock in the region, significantly reduced production across Guatemala, Honduras, Costa Rica, and neighbouring countries for several consecutive seasons and prompted large-scale replanting programmes with rust-resistant hybrid varieties (such as Catimor and other Robusta-Arabica hybrid lineages) in the years that followed.
- **Climate change and rust risk:** agricultural researchers and industry bodies have flagged warming temperatures at traditionally rust-marginal higher-altitude growing areas as a factor potentially expanding the geographic range in which rust can establish, an area of ongoing research and industry concern rather than a settled, quantified forecast.

5.3 Ethiopia and Origin Diversity

- Ethiopia, as the genetic birthplace of Arabica coffee, retains an unusually diverse range of heirloom varieties grown across a mix of wild forest, semi-forest, garden, and smallholder plantation systems. Ethiopian domestic consumption absorbs a substantial share of national production, and the country's production and export data infrastructure is generally less granular and timely than that available for Brazil or Colombia, making Ethiopian supply assessments somewhat less precise inputs to the global balance sheet than the corresponding Brazilian or Colombian data.

6. Demand Fundamentals

World coffee consumption is approximately 165–175 million bags per year and has grown steadily, driven by rising consumption in traditional importing markets and — increasingly — by growth in producing-country domestic consumption and in historically low-consumption emerging markets.

6.1 Consumption by Region

Region	Consumption (approx.)	Per-Capita Trend	Key Demand Driver	KC Curve Sensitivity
European Union	~40–45 M bags/yr	Mature, slow growth	Established consumption culture; specialty/premium segment growth; sustainability and traceability certification demand (EUDR — EU Deforestation Regulation — compliance)	Stable large demand base. EU import/compliance regulatory changes (EUDR) are a watched policy variable affecting origin sourcing patterns.
United States	~27–28 M bags/yr	Mature, modest growth	Specialty coffee and out-of-home (cafe) consumption growth; roaster and retail brand competition	Stable large demand base; US specialty segment growth supports premium-grade Arabica demand specifically.
Brazil (domestic)	~22–23 M bags/yr	Growing	Domestic consumption growth in the world's largest producing country; absorbs a significant share of Brazilian production directly	Reduces the exportable share of Brazilian production at the margin; a structurally growing domestic demand base in the dominant producing country.
Japan	~7–8 M bags/yr	Mature, stable	Established consumption culture; quality-focused demand	Stable; JPY exchange rate affects Japanese import cost and demand at the margin.
China	~3–4 M bags/yr (rapidly growing from small base)	Fast growth from a low base	Urbanisation; Western-style cafe chain expansion; younger consumer adoption of coffee culture	Currently small in absolute terms but the most-cited long-run emerging demand growth story for global coffee.
Other emerging markets (India, Southeast Asia, Middle East)	Variable, growing	Growing from low base	Urbanisation and rising incomes; cafe culture expansion	Collectively an important long-run demand growth layer, though individually smaller than the major established markets.

6.2 The Specialty Coffee Segment

- **Specialty coffee growth:** the specialty segment (higher-grade, often single-origin or traceable coffee, frequently commanding prices well above the KC futures-linked commodity grade) has grown significantly in major consuming markets over the past two decades, driven by cafe chain expansion, direct-trade and traceability marketing, and changing consumer preferences particularly among younger demographics.
- **Effect on KC fundamentals:** specialty-grade coffee is priced via direct relationships and quality-based differentials that sit above and somewhat independent of the KC futures price, though the futures price remains the reference point off which specialty differentials are typically negotiated. Specialty segment growth has, over the long run, supported overall Arabica demand

growth and modestly increased demand for the higher-grade end of the deliverable quality spectrum.

6.3 Out-of-Home Consumption and Economic Cycles

- Cafe and out-of-home coffee consumption (as distinct from at-home brewing) is more sensitive to discretionary consumer spending and economic cycles than at-home consumption, making employment and consumer confidence data relevant secondary indicators for the demand side of the coffee balance, particularly in mature markets with large cafe sectors.

7. Certified Stocks and the Physical Market

ICE-certified coffee stocks — green coffee held in ICE-licensed warehouses and graded as deliverable against the KC contract — are one of the most closely watched data series in the coffee market, serving a role analogous to LME on-warrant stocks in copper or COMEX registered stocks in gold.

7.1 ICE Certified Stocks

- **What the data shows:** ICE publishes daily certified stock data, broken down by origin/growth and warehouse location. Certified stocks represent coffee that has been graded and approved for delivery against the futures contract — a subset of total global green coffee inventory, not a measure of total world stocks.
- **The multi-year decline and subsequent volatility:** ICE certified coffee stocks fell to multi-decade lows at various points over the 2020s, a trend widely attributed to a combination of tight global supply-demand balances, the cost and logistics of certifying and holding stock against the contract relative to alternative uses of capital, and the specific Brazilian and Vietnamese supply disruptions discussed in earlier sections. Low certified stocks relative to open interest in nearby contracts have, at various points, been cited as a contributing factor to sharp front-month spread volatility.
- **Decertification and recertification flow:** coffee can be decertified (withdrawn from the certified pool, often for blending, re-grading, or sale into the physical/specialty market outside the futures delivery system) and new coffee can be certified as it passes grading. The flow rate of certifications and decertifications, not just the absolute stock level, provides additional context on physical market tightness.

7.2 ICO and Other Global Stock Estimates

- **International Coffee Organization (ICO):** publishes monthly composite price indicators and periodic global supply/demand/stock balance estimates, aggregating data from member producing and consuming countries. ICO data provides a broader, if less timely, context for total global stocks beyond the ICE-certified subset.
- **USDA Foreign Agricultural Service (FAS) coffee reports:** USDA publishes semi-annual (June and December) global coffee production, consumption, and stocks estimates covering all major producing and consuming countries, widely used by trade analysts as a comprehensive global balance sheet reference alongside ICO data.

7.3 The Roaster and Trade House Pipeline

- Coffee roasters typically maintain forward-purchased green coffee inventory covering some months of anticipated roasting requirements, hedged via a combination of KC futures, options, and fixed-price physical contracts with origin suppliers and trade houses. Roaster hedging and pricing decisions (particularly around the timing of fixing prices on previously-agreed differential contracts) contribute to futures market volume and open interest patterns, though the specific timing of individual roaster hedging activity is generally not separately published or directly observable.

8. The Forward Curve Structure — Zones and Spread Mechanics

Coffee trades in five delivery months per year (March, May, July, September, December), with December — the contract immediately following the Brazilian harvest's peak arrival into the export pipeline — typically the most liquid and closely watched.

8.1 Contract-by-Contract Fundamental Mapping

Contract	Reference Period	Primary Fundamental Driver	Secondary Driver	Key Signal
December (Z)	Prices the period immediately following the Brazilian harvest peak (Apr-Sep); the most liquid contract.	Brazilian harvest size and quality just confirmed; CONAB final-season estimates; Vietnamese harvest just beginning (Oct-Jan)	ICE certified stock trend; Colombian main-crop arrival (Oct-Dec)	CONAB final crop estimate; ICE certified stock level entering year-end
March (H)	Mid-marketing-year. Brazilian off-season (no Arabica harvest activity). Vietnamese harvest concluding.	Export pace from the just-completed Brazilian harvest vs USDA/ICO forecast; Brazilian frost risk window (Jun-Aug) approaching for the NEXT crop	Colombian mitaca secondary crop arrival (Apr-Jun, priced forward into this contract)	Weekly/monthly export data from Brazil and Colombia; early frost-season weather attention building
May (K)	Brazilian frost risk window is live (Jun-Aug spans into/around this contract). Flowering-period drought risk for the following year also relevant.	Real-time Brazilian winter weather (frost watch); Brazilian flowering conditions for the NEXT season's crop	Colombian mitaca crop confirmed; Central American harvest concluding	Daily/weekly Brazilian temperature forecasts during the frost-risk window; this is typically the most weather-headline-sensitive contract of the year
July (N)	Brazilian frost risk window largely behind (though late frosts are possible); early read on the new Brazilian crop forming.	Confirmed frost/drought damage assessment if an event occurred; early CONAB estimates for the upcoming harvest	Global roaster/consumption demand trends; Vietnamese new-crop planting outlook	CONAB first new-crop estimates; confirmed frost damage surveys if applicable
September (U)	Brazilian harvest underway and accelerating (Apr-Sep harvest window). New-crop size becoming clearer.	Brazilian harvest progress and quality; on-year/off-year biennial cycle position confirmed	Global stocks-to-consumption trajectory entering the new crop year	CONAB mid-harvest estimates; Brazilian export pace as new crop begins reaching port

8.2 Carry, Backwardation, and the Frost Premium

- **Normal carry:** in a well-supplied market, KC trades in a modest contango reflecting storage, financing, and insurance costs for holding physical green coffee (coffee stores considerably better than many soft commodities, as properly stored green coffee can remain gradeable for an extended period, though cup quality can deteriorate gradually over time — a slower-moving quality consideration than for, say, sugar or cocoa).

- **Backwardation as a tightness signal:** as with cocoa and sugar, KC backwardation (nearby trading above deferred) signals near-term physical scarcity — low certified stocks, strong immediate roaster demand, or a confirmed recent supply disruption (frost or drought damage just confirmed).
- **The "frost premium" as a distinct curve phenomenon:** in years where Brazilian winter weather forecasts raise frost concern without (yet) confirmed damage, the contracts spanning the risk window (typically May-July) can carry an embedded weather-risk premium relative to contracts on either side of the window, a temporary curve distortion that compresses rapidly once the frost risk period passes without confirmed damage, or expands sharply if damage is confirmed.

8.3 The Arabica-Robusta Spread as a Distinct Tradeable Relationship

- The spread between KC (Arabica, ICE US, ¢/lb) and Robusta (ICE Europe, USD/tonne) requires unit conversion but is closely tracked as a relative-value signal: a widening Arabica premium (beyond historical norms) signals either Arabica-specific tightness (Brazilian frost/drought) or Robusta-specific abundance (strong Vietnamese harvest), and vice versa for a narrowing premium. The spread is also a leading indicator for blend substitution dynamics described in Section 1.2.

9. Weather, ENSO, and Production Risk

9.1 ENSO and Coffee Production

Country	El Nino Effect	La Nina Effect	Typical Lead Time	KC Impact
Brazil	Generally associated with reduced rainfall in parts of the growing region during the season; effect on frost risk itself is less directly ENSO-linked than the drought channel.	Some La Nina years have been associated with cooler polar air incursions reaching further north into Brazil, a factor monitored (though not a precise forecasting rule) in assessing winter frost risk.	Seasonal (winter frost window; flowering-period drought risk)	Brazil's weather risk (frost specifically) is driven by specific synoptic weather patterns during the Jun-Aug window more directly than by the ENSO state itself, though ENSO contributes to the broader seasonal rainfall/temperature backdrop.
Vietnam	Drought stress; historically associated with reduced Vietnamese Robusta yields.	Generally more favourable rainfall for Vietnamese Robusta.	6-9 months (growing season)	A weak Vietnamese crop in an El Nino year supports the Robusta price and narrows the Arabica-Robusta spread, all else equal.
Colombia	Drought stress.	Excess rainfall; associated with leaf rust (roya) outbreak risk as described in Section 5.1.	Current and following season	Both ENSO extremes carry some Colombian production risk via different mechanisms (drought vs disease).
Central America	Drought stress in parts of the region.	Excess rainfall and elevated rust risk, as in Colombia.	Current and following season	Regional production risk; relevant to the washed-Arabica supply pool that competes with/complements Colombian and Brazilian washed coffee.
Indonesia	Drought stress (especially Sumatra/Java Robusta and Arabica areas).	Generally more favourable.	Seasonal	Secondary Robusta supply effect; smaller than the Vietnam relationship but directionally similar.

9.2 Brazilian Frost Forecasting — The Specialised Weather Signal

- **Synoptic pattern monitoring:** Brazilian frost risk is assessed through monitoring of polar air mass incursions (locally and meteorologically referred to via terms such as "friagem" events) pushing northward from Argentina/the South Atlantic into southern Brazil during the June–August window. Commercial weather services and Brazilian meteorological agencies (INMET) provide forecasts with actionable lead times typically in the range of several days to roughly two weeks, meaning frost risk is generally not a multi-month-ahead forecastable event in the way that, for example, an ENSO-driven monsoon forecast can provide several months of lead time for Indian agriculture.
- **The "frost season" as a calendar-defined watch period:** the market enters a heightened state of attentiveness to Brazilian temperature forecasts every June through August regardless of the specific forecast at any given moment, simply because this is the calendar window in which frost is physically possible — a recurring seasonal feature of trader attention distinct from a forecast-driven signal itself.
- **Distinguishing a genuine frost event from a false alarm:** forecast models can indicate elevated frost risk that does not materialise, or that affects a smaller geographic area or a less significant growing region than initially feared. The market has, at various points, priced in meaningful risk premium around forecast frost events that ultimately produced limited confirmed

damage, underscoring that the priced-in risk premium during the watch window and the eventual confirmed damage are two distinct things that can diverge significantly.

10. Macro, Currency, and Cross-Market Drivers

10.1 Brazilian Real and Colombian/Vietnamese Currencies

- As described in Section 3.4, BRL/USD is the primary currency variable, operating through the same mechanism as in Brazilian sugar: BRL depreciation improves farmer economics in local-currency terms at any given USD coffee price, incentivising selling and structurally pressuring USD-denominated KC prices; BRL appreciation has the opposite effect.
- Colombian Peso (COP/USD) and Vietnamese Dong (VND/USD) operate analogously for their respective countries' farmer/exporter economics, though with smaller aggregate market impact given Brazil's outsized share of global production.

10.2 Freight and Logistics Costs

- Coffee is a bulky, moderate-value-density commodity for which ocean freight costs (from origin ports such as Santos, Brazil; Buenaventura, Colombia; and Ho Chi Minh City, Vietnam, to consuming-country ports) represent a meaningful component of landed cost. Periods of elevated global container and bulk shipping rates (such as those experienced during the 2021–2022 global shipping disruption period) can widen origin-to-destination basis and affect the relative cost competitiveness of different origins reaching a given consuming market.

10.3 Speculative Positioning — COT Data

- **CFTC COT for Coffee C:** weekly managed money net position data is closely tracked, as coffee is among the more heavily speculated soft commodity futures. Extreme net long or net short positioning is generally treated as a crowding/reversal-risk indicator, consistent with the positioning framework used across the other soft commodity markets in this document series, rather than as a standalone directional signal.
- **Commercial (trade house and roaster) positioning:** large physical trade houses and roasters use KC futures and options to hedge forward physical purchase and sale commitments; the scale and direction of commercial positioning provides context for interpreting the managed money speculative positioning data.

10.4 Sustainability and Regulatory Developments

- **EU Deforestation Regulation (EUDR):** EU import requirements concerning deforestation-free supply chain documentation for coffee (among other commodities) have introduced a compliance and traceability dimension to coffee sourcing for exporters targeting the EU market, a regulatory development that trade participants monitor for its potential effect on origin sourcing patterns and compliance costs, particularly for smallholder-dominated origins where traceability documentation infrastructure is less developed.
- **Certification schemes (Fairtrade, Rainforest Alliance, organic):** voluntary certification programmes affect the specialty and premium segment of the market discussed in Section 6.2, creating differentiated pricing for certified-compliant coffee that sits alongside, but does not replace, the KC futures price as the underlying commodity reference.

11. Documented Seasonal Patterns

Coffee's seasonal pattern is dominated by the Brazilian crop calendar, with the May-July frost risk window standing out as the single most distinctive recurring feature of the annual KC trading calendar.

Period	Brazilian Crop Stage	Typical KC Behaviour	Key Variables	Reliability
Jan-Mar (off-season)	No Arabica harvest activity (post-harvest marketing continues). Vietnamese harvest concluding.	Generally calmer trading; driven by export pace data and carry-over stock assessment rather than acute weather risk.	Brazilian/Colombian export data; ICE certified stock trend; Vietnamese crop-end estimates	Moderate
Apr-May (flowering risk + early frost window)	Brazilian flowering period for the FOLLOWING year's crop (forward-looking drought risk); frost risk window opening.	Attention begins shifting to Brazilian winter weather forecasts. Flowering-period drought reports begin to affect deferred contracts.	Brazilian rainfall during flowering; early-season frost forecast model output	Moderate, rising into the window
Jun-Aug (frost risk window)	The defining annual weather-risk period. Live frost risk in southern growing regions.	The most weather-headline-sensitive period of the year. Sharp, rapid price moves possible on forecast shifts and (if applicable) confirmed damage reports.	Daily/weekly Brazilian temperature forecasts; INMET and commercial weather service updates; confirmed damage assessments if an event occurs	High — the window itself is calendar-certain even though any specific event is not
Sep-Oct (post-frost-window, harvest approaching)	Frost risk has largely passed (though late events remain possible). Early new-crop size assessment forming.	Market transitions from weather-risk pricing to crop-size assessment. CONAB and other early estimates become the focus.	CONAB first new-crop estimates; confirmed damage surveys if a frost event occurred over the summer	Moderate
Nov-Dec (harvest underway)	Brazilian Arabica harvest in progress and accelerating toward peak.	Harvest progress and quality data refine the crop-size picture; December contract (the most liquid) reflects the culmination of the season's information.	CONAB mid/late-harvest estimates; export pace as new crop reaches port	Moderate-High

11.1 Key Documented Historical Events

- **The 1975 Brazilian "Black Frost":** as described in Section 3.2, the most severe documented Brazilian coffee frost event, producing a multi-year supply disruption as damaged trees required years to replace and return to full productive capacity, and a historic price spike that remains the reference event for the entire frost-risk framework described throughout this document.
- **The 2012-2013 Central American leaf rust epidemic:** a severe, multi-country roya outbreak that reduced Central American Arabica production for several consecutive seasons and prompted large-scale replanting with rust-resistant varieties, as described in Section 5.2.
- **July 2021 Brazilian frost scare:** a frost event affecting parts of Minas Gerais and Parana in July 2021, occurring against an already-tightening global supply backdrop (the 2021 crop was also an off-year in the biennial cycle and faced earlier-season drought stress), contributing to a sharp KC

rally as the combination of confirmed frost damage, biennial off-year timing, and prior drought stress compounded into a multi-factor supply shock — illustrating the interaction effects described in Section 2.1.

- **2023-2024 Vietnamese Robusta shortage and the Arabica-Robusta spread:** a period of Vietnamese drought stress (linked to El Nino conditions) reducing Robusta availability coincided with strong global demand, producing record or near-record Robusta prices and a notable narrowing of the historical Arabica premium over Robusta, illustrating the Arabica-Robusta substitution dynamics described in Section 1.2 operating in practice.

12. Key Data Sources and Release Schedule

Report	Publisher	Frequency	Typical Timing	Primary KC Curve Impact
CONAB Brazilian coffee crop estimates	CONAB (Brazil)	Several times per year	Varies (Jan, May, Sep typical windows)	All contracts — the most important single-country production data series; precision increases as harvest approaches
INMET / commercial weather service Brazilian forecasts	INMET / DTN / Maxar / others	Daily during frost-risk season	Jun-Aug	Front-to-mid contracts — the most acute, fastest-moving signal in the entire coffee market during the frost window
USDA FAS Coffee: World Markets and Trade	USDA FAS	Semi-annual	June and December	All contracts — comprehensive global production/consumption/stocks balance sheet
ICO Coffee Market Report	International Coffee Organization	Monthly	~Mid-month	All contracts — composite price indicators and periodic global balance updates
FNC (Colombian Coffee Growers Federation) production/export data	FNC	Monthly	~1st week of month	C2-C5 — Colombian supply confirmation; washed-Arabica premium signal
Vietnam Customs / industry association export data	Vietnam Customs / VICOFA	Monthly	Variable	Arabica-Robusta spread — Vietnamese Robusta export pace signal
ICE certified coffee stocks	ICE Futures US	Daily	After close	Front contract — physical tightness signal; delivery squeeze risk assessment
NOAA ENSO forecast update	NOAA CPC / IRI	Monthly	~Mid-month	C4-C8 — seasonal rainfall probability for Vietnam, Colombia, Indonesia
BRL/USD exchange rate	Reuters / Bloomberg / BCB	Daily / Continuous	Market hours	All contracts — Brazilian farmer selling economics; continuously relevant
CFTC Commitments of Traders	CFTC	Weekly	Fri ~15:30 ET (3-day lag)	Positioning / crowding — all contracts; extreme managed money positions as contrary indicator
Global container/bulk freight rate indices	Freightos, Baltic Exchange, others	Weekly	Variable	Basis — origin-to-destination landed cost competitiveness across origins
EU Deforestation Regulation (EUDR) compliance deadlines/updates	European Commission	As scheduled/ amended	Variable	Back curve — origin sourcing pattern and compliance cost implications for EU-bound coffee

ICE Coffee C is, more than almost any other soft commodity, a single-country, single-season weather-risk market layered onto a structurally biennial production cycle. Brazil's dual exposure to genuinely unusual frost risk (a hazard with no parallel among other major tropical commodities) and to drought during its critical flowering and cherry-development windows, combined with the multi-year recovery timeline that follows tree-damaging frost events specifically, gives the Brazilian winter

(June-August) an outsized and calendar-certain role in the annual KC trading cycle that has no equivalent in the other soft commodities covered elsewhere in this document series.

For spread traders, the most reliable edges come from three recurring patterns: (1) the May-July frost-risk premium build and decay — contracts spanning the Brazilian winter carry an embedded weather premium that compresses rapidly once the season passes without confirmed damage and expands sharply if damage is confirmed, with the speed of repricing in either direction creating distinct tactical opportunities; (2) the biennial bearing cycle interaction with weather — a weather shock landing in a scheduled Brazilian off-year compounds into a larger price reaction than the same weather event would produce in an on-year, making knowledge of where Brazil sits in its biennial cycle an essential input to weather-event sizing; and (3) the Arabica-Robusta spread as a substitution-driven relative value trade — tracking Vietnamese and Brazilian Robusta (Conilon) conditions alongside Brazilian Arabica conditions captures the blending and instant-coffee substitution dynamics that connect the two markets.